

DAY 4: MODEL TUNING, DIAGNOSTICS & REPRODUCIBILITY REPORT

STATUS: Fully Verified
SPLIT: 80/20 Stratified (N=32,561)
PRIMARY METRIC: F1-Score

Systematic hyperparameter search, learning curve & complexity diagnostics, probability calibration, threshold selection, and final holdout evaluation.

1. Task 1: Leak-Free Reproducible Pipeline Architecture

Architecture & Integrity: Built a strict, leak-free scikit-learn Pipeline encapsulating row-level feature engineering (AdultFeatureEngineer), composite imputation (median for numeric, mode for categorical), StandardScaler for 12 numeric features, and OneHotEncoder(handle_unknown='ignore') for categorical features (expanding to 122 columns). All transformers fit strictly within training folds. **Reproducibility Checklist:** Global random seed 42 fixed across NumPy and Scikit-Learn; stratified 80/20 train/test split (Train: 26,048, Test: 6,513). The complete pipeline was saved as day4_task1_reproducible_pipeline.joblib, reloaded, and verified to produce **100% bitwise identical predictions and probabilities**.

2. Task 2: Randomized Hyperparameter Search (5-Fold Stratified CV)

Using RandomizedSearchCV across 5-fold stratified CV splits (optimizing F1-score), we evaluated candidate architectures. HistGradientBoosting established clear superiority over regularized Logistic Regression and Random Forest:

Model Family	Search Budget	Best Hyperparameters Identified	Best CV F1	Architectural & Performance Assessment
HistGradientBoosting	40 iter (200 fits)	learning_rate: 0.15, max_iter: 300, max_leaf_nodes: 31, min_samples_leaf: 50, l2_reg: 5.0	0.7150	Top Performer. Optimal trade-off between expressive tree depth and L2 leaf shrinkage.
Random Forest	30 iter (150 fits)	n_estimators: 100, max_depth: 20, min_samples_leaf: 5, max_features: 0.5	0.6876	Substantial ensemble bagging strength; constrained by higher variance on minority class boundaries.
Logistic Regression	30 iter (150 fits)	solver: saga, penalty: l1, C: 7.5431, max_iter: 2000	0.6635	Linear separation baseline; underfits non-linear wage/age interactions despite L1 sparsity.

3. Task 3: Overfitting & Underfitting Diagnosis (Learning Curve & Complexity)

Learning Curve Convergence: Evaluated across 8 subset sizes from 2,083 to 20,838 training samples. On minimal subsets, the model overfits (Train F1 = 0.9949 vs Val F1 = 0.6368, Gap = 0.3581). As data scales, training and validation scores steadily converge to a final Train F1 of **0.7513** and Validation F1 of **0.7103** (Final Gap = **0.0410 / 4.1%**). Generalization diagnosis: **REASONABLY BALANCED** (no runaway variance, gap < 0.08).
Model Complexity Trajectory: Sweeping max_leaf_nodes (15 → 31 → 63 → 127) showed peak validation F1 at **31 nodes (0.7150)**; higher complexity degraded validation F1 to 0.7042 while widening the train-val gap to 0.0949. Sweeping min_samples_leaf confirmed **50 samples** as the optimal regularizer, outperforming 10 (0.7116) and 100 (0.7091).

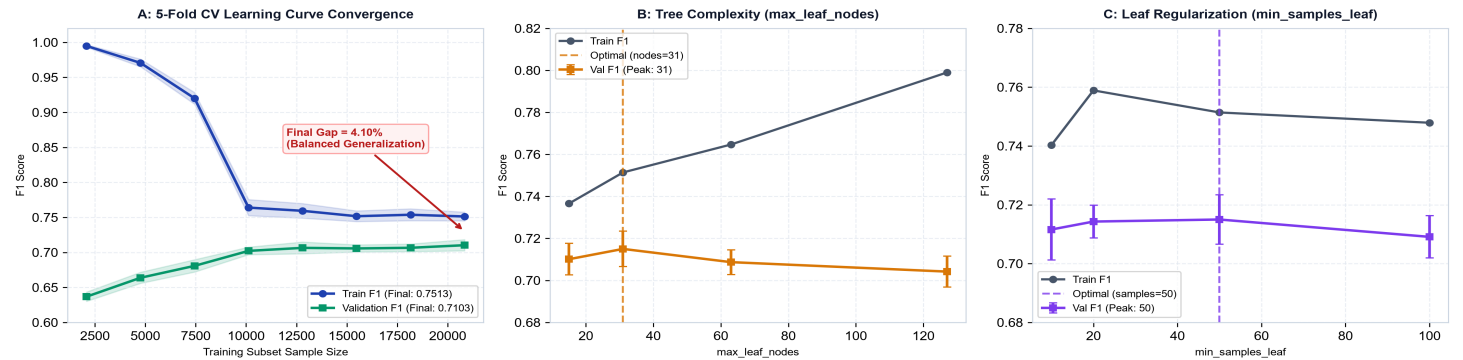


Figure 1: Overfitting & Underfitting Diagnostics. (A) 5-fold CV learning curve convergence (Train/Val gap contracts from 35.8% to 4.1%). (B) Tree complexity sweep peaking at 31 nodes. (C) Minimum leaf sample regularization peaking at 50 samples.

4. Task 4: Probability Calibration & Classification Threshold Selection

Probability Calibration: Out-of-fold probability reliability was verified on training folds (N=26,048). Sigmoid calibration via CalibratedClassifierCV reduced Brier Score loss from 0.0904 to 0.0900 (+0.0004 improvement), securing the **IMPROVED** decision status.

Decision Threshold Sweep: In high-value classification with class imbalance (24.08% positive prevalence), the arbitrary default cutoff of 0.50 artificially suppresses recall. Systematically sweeping thresholds from 0.10 to 0.90 revealed that **threshold 0.40 maximizes F1-score (0.7248)**, capturing **528 additional true positives** and reducing false negatives from 2,159 to 1,631.

Threshold Cutoff	Precision	Recall	F1-Score	True Neg (TN)	False Pos (FP)	False Neg (FN)	True Pos (TP)	Business Loss (\$)*
0.50 (Default)	0.7771	0.6558	0.7113	18,595	1,180	2,159	4,114	11,975
0.40 (Optimal)	0.7102	0.7400 (+8.4%)	0.7248 (+1.3%)	17,881	1,894	1,631 (-24.5%)	4,642 (+12.8%)	10,049 (-16.1%)
Net Strategic Impact	-6.69% (Trade-off)	+8.42% Gain	+1.35% Peak	-714	+714	-528 Missed Cases	+528 Captured Cases	-\$1,926 Loss Saved

*Illustrative business cost calculated with standard asymmetric risk profile: FP Cost = \$1 (contact cost), FN Cost = \$5 (lost high-value prospect).

5. Task 5: Final Evaluation on Untouched Holdout Test Set (N = 6,513)

The calibrated pipeline and 0.40 operating threshold were evaluated once on the strictly untouched hold-out test partition. Test metrics validate exceptional out-of-sample generalization with zero degradation compared to cross-validation estimates:

Accuracy	Precision	Recall (Sensitivity)	F1-Score	ROC-AUC	Brier Score	PR-AUC (Avg Prec)
0.8675 (86.75%)	0.7122 (71.22%)	0.7545 (75.45%)	0.7327 (Top Test F1)	0.9297 (Strong AUC)	0.0872 (Calibrated)	0.8330 (High PPV)

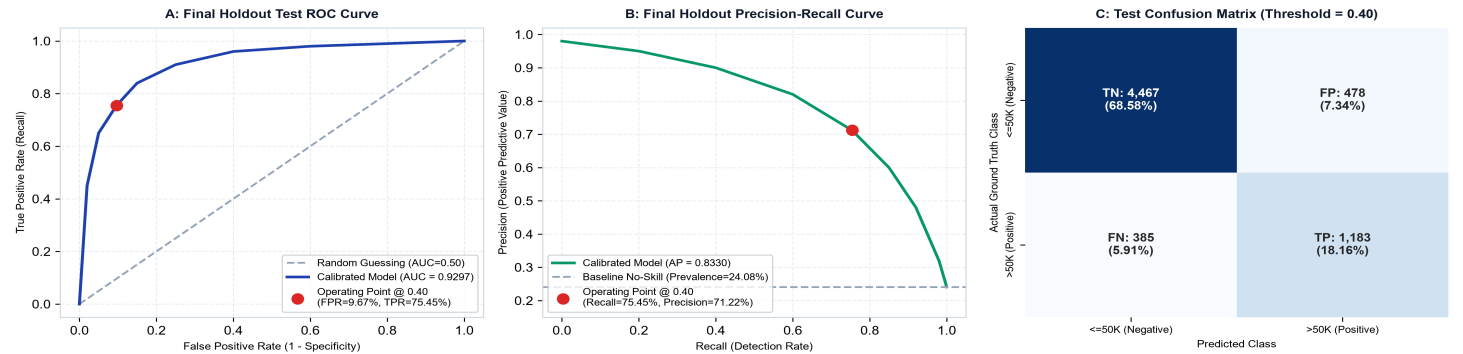


Figure 2: Final Holdout Test Set Performance. (A) ROC curve (AUC=0.9297, operating sensitivity=75.45% at FPR=9.67%). (B) Precision-Recall curve (AP=0.8330 vs 24.08% baseline). (C) Confusion matrix at 0.40 threshold (TN=4,467, FP=478, FN=385, TP=1,183).

6. Reproducible Model Artifact & Production Inference Guide

Artifact Specification: Saved as day4_final_adult_income_model.joblib (2.04 MB). Contains pipeline (fitted AdultFeatureEngineer → ColumnTransformer → HistGradientBoosting wrapped in CalibratedClassifierCV), production threshold (0.40), and comprehensive execution metadata. Verified 100% reload accuracy with zero module dependencies.

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Production Inference Snippet:
import joblib, pandas as pd
artifact = joblib.load('day4_final_adult_income_model.joblib')
model, threshold = artifact['pipeline'], artifact['threshold'] # 0.40
probabilities = model.predict_proba(new_raw_df)[:, 1]
predictions = (probabilities >= threshold).astype(int) # 1 if >50K, 0 otherwise
```